

Advanced Statistical Data Analysis

Worksheet Week 8

Exercise 1: Ship Damages Data (Data set `ships` in the package `MASS`)

- a) Repeat the analysis for the Haverien data that was shown in class and perform a residual analysis.
- b) Repeat the analysis of this dataset by assuming that the response variable is “the number of damage” ($= Y$) and the variable `service` is an ordinary explanatory variable.

Does the data analysis indicate that this model does not fit, or does it indicate that a rate model could describe the data more adequately?

Exercise 2: Repeat the analysis for the Dial-A-Ride data that was shown in class. (Data set `Dial-a-ride.dat`).

Exercise 3: Dataset about the number of fractures (Y) that occur in the upper seams of coal mines in the Appalachian region of western Virginia. (See also Exercise 2 on Worksheet-W7)

- a) Is it also possible to use a rate model?
- b) Under which circumstances would this approach be appropriate?